

MOHGCN: A trustworthy multi-omics data integration framework based on specificity-aware heterogeneous graph convolutional neural networks for disease diagnosis

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ABSTRACT

With the advancement of cutting-edge sequencing methodologies, the integration of multi-omics data provides invaluable opportunities for researchers to study complex diseases from a molecular perspective while at the same time being challenged by the deployment of safety-critical applications such as computer-aided diagnostics. However, existing methods in multi-omics data integration primarily explore interactions between omics or samples, neglecting high-order interaction information among biomolecules specific to certain diseases. In this work, we propose MOHGCN, a trustworthy multi-omics data integration framework based on specificity-aware heterogeneous graph convolutional neural networks for disease diagnosis, aiming to maximize the utilization of biomolecular interactions in patients with specific diseases for precise diagnosis to enhance the model's credibility. In the approach, we constructed a heterogeneous graph of samples and genes and devised the HGCN graph convolution model specifically tailored to the sample-gene heterogeneous graph. Concurrently, techniques such as trustworthy attention weights and self-attention mechanisms were incorporated to unveil relationships between different omics, facilitating the efficient integration of multi-omics data. Through comprehensive experimentation on four publicly available multi-omics medical datasets, our proposed framework consistently demonstrates superior performance across various classification tasks. Simultaneously, the experimental results substantiate the model's effectiveness in extracting features from multi-omics data and unveiling latent associations among different omics.

1. Introduction

Advancements in sequencing technologies have generated massive multi-omics data for various complex diseases, offering researchers valuable insights into these diseases at the molecular level (Boehm, Khosravi, Vanguri, Gao, & Shah, 2022; Rhodes, Sweatt, & Maron, 2022). Integrating such data across different omics types is pivotal for elucidating pathogenic mechanisms and advancing precise diagnostics and treatment options for patients (He, Liu, Zuo, Shi, & Jing, 2023; Jiang et al., 2022; Pérez-Villa et al., 2023). The successful application of computational technologies in fields like computer-aided diagnosis further highlights their strength in handling and analyzing complex datasets (Borlea, Precup, & Borlea, 2022; Ciecholewski, 2011), and underscores their potential in integrating multi-omics data for precise disease diagnostics, a prominent focus in current research (Ma et al.,

2020; Yang & Wu, 2021). Despite advancements in biomedicine, existing methods for multi-omics data classification are challenged by high-dimensionality, noise, and limited sample sizes. These limitations pose substantial challenges to the deployment of computer-aided diagnosis in safety-critical applications. Hence, a reliable framework for the integration of multi-omics data is of paramount importance for practical applications.

For the multi-omics data integration framework, conventional approaches have predominantly concentrated on the straightforward concatenation of diverse omics data (Chang et al., 2018; Joshi, Rienks, Theofilatos, & Mayr, 2021; Lanckriet, De Bie, Cristianini, Jordan, & Noble, 2004; Ronen, Hayat, & Akalin, 2019; Tarazona, Arzalluz-Luque,

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& Conesa, 2021). While these approaches can yield relatively straightforward embeddings of data, they overlook the correlation and redundancy among different types of omics data, failing to comprehensively and effectively capture the biological processes. In recent years, Graph Neural Networks (GNNs) (Gilmer, Schoenholz, Riley, Vinyals, & Dahl, 2017; Liu et al., 2021; Wu et al., 2020; Zhang, Tong, Xu, & Maciejewski, 2019), by leveraging the intricate interactions among samples under different omics data, have offered a new perspective to integrate multi-omics data (Wang et al., 2021; Zheng, Tang, Wan, Hu and Zhang, 2023). Although these methods have achieved considerable success, existing approaches have not taken into account the high-order interaction information genuinely existing among biomolecules specific to certain diseases. This limitation hinders the model's ability to ensure stability and interpretability. Interactions among biomolecules under specific diseases often better reflect the molecular mechanisms within patients, thereby enhancing the model's credibility (Wu, Zhou, Mu, & Hu, 2022). Consequently, it is necessary and crucial to reconstruct the interactions among biomolecules under specific diseases and integrate them into the learning process of multi-omics data.

In this work, we present a trustworthy multi-omics data integration framework, MOHGCN, for disease diagnosis based on a Specificity-Aware Heterogeneous Graph Convolutional Neural Network. This framework models the interactions between biomolecules specific to certain diseases across different omics while investigating the relationships among various omics types to facilitate the efficient integration of multi-omics data. Specifically, we build heterogeneous graphs depicting interactions between samples and biomolecules under different omics types. And the Specificity-Aware Heterogeneous Graph Convolutional Networks (HGCNs) are designed for different omics types to extract features for samples. Then, a trustworthy attention weight technique is introduced to enhance the confidence of feature extraction by aligning the decision-making process of each omics type with the true labels. Finally, through a multi-head self-attention mechanism, we thoroughly explore potential relationships between features from different omics types, achieving efficient integration of multi-omics data.

The contributions of this work can be summarized as follows:

1. We propose a trustworthy multi-omics data integration framework, MOHGCN, which fully considers the sample-biomolecule interactions in a specific disease and designed a new convolutional module of HGCNs to improve the interpretability of the model;
2. We designed a trustworthy cross-modal feature fusion and handling method based on the self-attention mechanism and true class probability (TCP) (Corbière, Thome, Bar-Hen, Cord, & Pérez, 2019) to fully exploit the correlations between different omics and improve the classification ability of the model;
3. We conducted extensive experiments on four real-world multi-omics medical datasets to validate the effectiveness of the proposed multi-omics data integration framework, MOHGCN.

2. Related work

2.1. Application of computer technology for computer-aided diagnosis

With the rapid advancements in computer science, computational technologies have demonstrated their formidable capabilities in handling and analyzing complex data across various fields. For instance, the Support Vector Machine (SVM) method has been successfully applied to identify patterns in gallbladder ultrasound images (Ciecholewski, 2011). Similarly, enhanced K-means algorithms have shown significant success in collaborative decision-making for big data (Borlea et al., 2022), and XOR-based techniques have proven effective in network anomaly detection (Protic & Stankovic, 2023).

The advent of cutting-edge sequencing technologies has introduced a wealth of multi-omics data (genomic, transcriptomic, and proteomic) from patients with complex diseases. This abundance of diverse and intricate data presents a unique opportunity for precise diagnosis through

the integration of multi-omics data. Effectively merging these heterogeneous data sources to enhance the predictive power and accuracy of classification models, thereby accurately categorizing patients into different disease states or subtypes, has become a research focal point (Ma et al., 2020; Yang & Wu, 2021). For example, Ma et al. (2020) developed an XGBoost-based approach to integrate multi-omics data for precise cancer patient diagnosis. Yang and Wu (2021) designed a diagnostic model for lung cancer subtypes that combines Random Forest (RF) and Support Vector Machine (SVM) methods.

Although traditional machine learning approaches have achieved certain success in computer-aided diagnosis, they face significant challenges when dealing with the high-dimensionality, high noise levels, and low sample sizes characteristic of multi-omics data. These limitations often render conventional methods less effective in addressing the complexities inherent in such datasets. Recently, the rise of deep learning technologies has provided new perspectives and tools for integrating and analyzing multi-omics data, offering potential solutions to overcome these challenges.

2.2. Deep learning methods for multi-omics data integration

The rapid advancement of computer technologies, particularly in deep learning, has turned utilizing neural networks (NN) for the integration of multi-omics data into a burgeoning research area (Joshi et al., 2021; Tarazona et al., 2021). Early methods for multi-omics data integration involved the simple concatenation of features obtained from different omics data to derive joint representations. Neural networks were subsequently employed to explore these joint embeddings (Chang et al., 2018; Lanckriet et al., 2004; Ronen et al., 2019). For instance, Lanckriet et al. (2004) integrated multiple omics information, including amino acid sequence data and gene expression data, using SVM for specific protein recognition. Ronen et al. (2019) fed concatenated multi-omics data into Stacked Variational Autoencoders (SVAE) to obtain joint embeddings for different omics types. However, these early fusion strategies did not adequately consider the correlations between different types of omics data, making it challenging to handle heterogeneous or inconsistent multi-omics data effectively.

To comprehensively investigate the correlations among various types of omics data, several methods for multi-omics data fusion have been proposed (Han, Yang, Huang, Zhang & Yao, 2022; Singh et al., 2019; Van De Wiel, Lien, Verlaet, van Wieringen, & Wilting, 2016). For example, an adaptive group-regularized ridge regression method was proposed by Van De Wiel et al. (2016), employing adaptive group regularization to integrate methylation microarray and annotation data for patient diagnostic classification. Singh et al. (2019) proposed the utilization of Data Integration Analysis for Biomarker Discovery using Latent components (DIABLO), a method capable of identifying common information across multiple omics types and achieving cancer subtype recognition. Despite the notable advancements achieved by these methods for integrating multi-omics data in comparison to early fusion approaches, they tend to neglect potential high-order interactions among samples and biomolecules. The consideration of such high-order information is critical for effectively learning from the features present in multi-omics data (Lovino et al., 2022; Zheng, Liu et al., 2023). The advancements in Graph Neural Networks (GNNs) (Gilmer et al., 2017; Liu et al., 2021; Wu et al., 2020; Zhang et al., 2019) have paved the way for researchers to explore new perspectives.

2.3. GNNs for multi-omics data analysis

GNNs iteratively update node embeddings by aggregating information from their neighborhoods, enabling the integration of high-order structural information across multiple layers (Gilmer et al., 2017). Consequently, numerous methods based on GNNs have emerged for the analysis of multi-omics data (Wang et al., 2021; Zheng, Tang et al., 2023). For instance, to leverage correlations between patients,

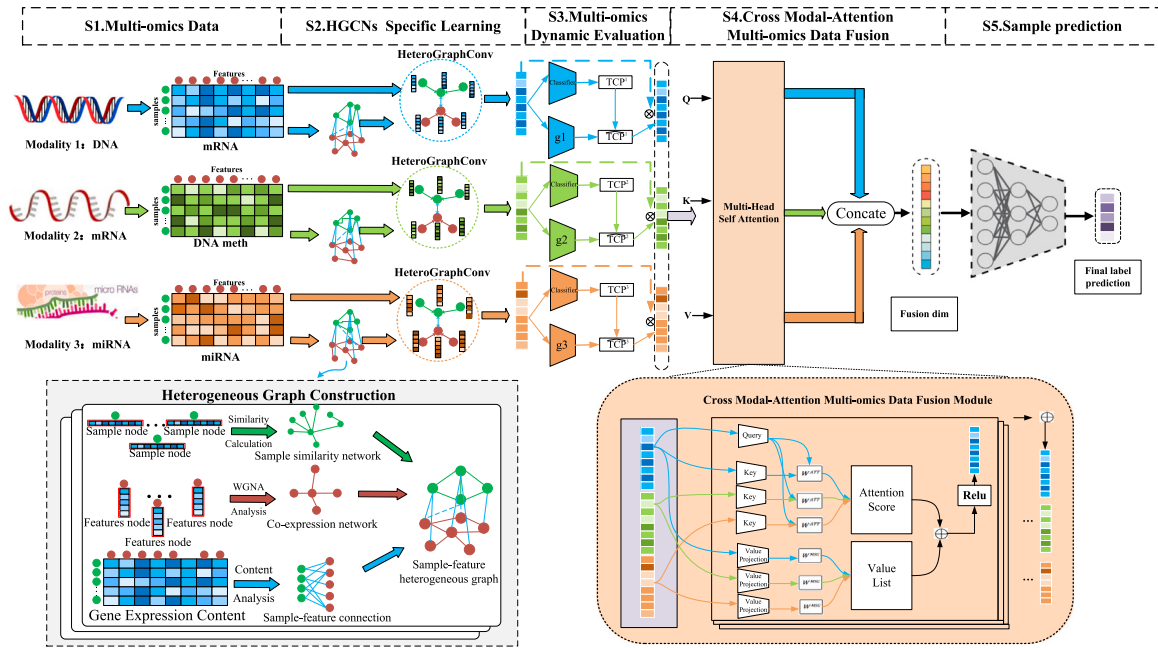


Fig. 1. Overview of MOHGCN. S1. Sample multi-omics data input. S2. HGCNs omics-specific feature extraction module. S3. Multi-omics dynamic evaluation module. S4. Cross modal-attention multi-omics data fusion module. S5. Fully connected neural network sample prediction module.

MOGONET was introduced by Wang et al. (2021). This method employs GNN to analyze the patient similarity graph derived from multi-omics data, facilitating biomedical data classification. Additionally, a reliable multi-omics classification network, MLCLNet, was proposed by Zheng, Tang et al. (2023). This network employs distinct Graph Convolutional Networks (GCNs) trained for different omics types to thoroughly extract features from various multi-omics data for subsequent patient classification.

Owing to the interplay among genes in modulating disease progression, a rational approach involves the transformation of multi-omics data into a graph format. In this configuration, individual genes assume the role of nodes, while their interactions are delineated as edges. For example, Rhee, Seo, and Kim (2018) employed the STRING database graph (Szklarczyk et al., 2015) with Graph Neural Networks (GNN) and deep feedforward networks to extract graph features for predicting breast cancer. In another study, Kong and Yu (2018) utilized the HINT database (Das & Yu, 2012) to construct a gene graph. They utilized a deep feedforward network for graph embedding to predict disease outcomes. While these approaches (Kong & Yu, 2018; Rhee et al., 2018) make use of knowledge about relationships among genes, it is noteworthy that HINT and STRING represent general protein interaction graphs and may not precisely capture gene interactions specific to a particular disease.

In this work, we propose a trustworthy multi-omics data integration framework based on Specificity-Aware Heterogeneous Graph Convolutional Neural Networks. Taking into account the interaction information among samples and genes under each omics type for a specific disease, we design convolutional modules on heterogeneous graphs to thoroughly extract features. The framework aims to make full use of the interactions between sample biomolecules under a specific disease for feature extraction. It also seeks to reveal the relationships between different omics types to better accomplish the ultimate task of sample classification. In the subsequent section, a comprehensive elucidation of MOHGCN will be presented.

3. Method

3.1. Overview of MOHGCN

In this work, we present a credible framework for integrating multi-omics data, MOHGCN, based on Specificity-Aware Heterogeneous Graph Convolutional Neural Networks, designed for disease diagnosis, as depicted in Fig. 1. To effectively leverage associations between samples and biomolecules and thoroughly explore specificity across different omics types, MOHGCN incorporates various technologies, including heterogeneous graph feature extraction, adaptive classification confidence assessment, and cross-modal feature fusion. The framework comprises four main steps: **1. Construction of Heterogeneous Graph for Sample-Genes Interaction:** A heterogeneous graph is constructed for samples and their biological features, including mRNA/DNA methylation/miRNA. This graph structure facilitates the propagation of features through both self-connections and connections with associated genes. **2. Omics-Specific Feature Extraction using Heterogeneous Graph Convolutional Networks (HGCN):** Utilizing the heterogeneous graph and omics features, a heterogeneous graph convolutional network (HGCN) is employed for each omics data type. Each HGCN is trained with real labels, effectively enhancing the model's ability to aggregate sample information within each omics type. **3. Confidence Learning Module for Omics features:** To address the imbalance caused by the heterogeneity of multi-omics data, an adaptive confidence attention weighting technique is introduced. Leveraging True-Class-Probability (TCP) (Corbière et al., 2019), this approach aligns the decision-making process of each omics type with the true labels, thereby enhancing the credibility of the model's classification. **4. Cross Modal-Attention Multi-omics Data Fusion:** Through a multi-head self-attention mechanism in the fourth step, thorough exploration of correlations between different omics types is achieved. The attention scores obtained are then used to update the features of different omics types. Finally, the features from different omics types are concatenated and input into a fully connected neural network to obtain the ultimate category predictions.

3.2. Heterogeneous graph construction

In the MOHGCN framework, a sample-gene heterogeneous graph is formulated for each omics type. Biomolecular functional interaction

networks are important for elucidating the pathogenesis of complex diseases (Lu et al., 2020). Heterogeneous graphs, adept at capturing intrinsic topological features of omics data, can enhance a holistic comprehension of molecular interactions within biological systems, thereby providing complementary information for sample classification from various perspectives (Nazem-Bokaee, Hom, Warden, Mathews, & Gueidan, 2021). A heterogeneous graph for samples and associated genes (mRNA/DNA methylation/miRNA) was constructed, comprising a similarity network among samples, a co-expression network among genes, and a sample–gene association network. This graph structure design ensures proximity of similar sample nodes, aiding in sample category differentiation. Additionally, it allows samples to propagate features not only through self-association to adjacent feature nodes but also through their associated genes.

Formally, the entire heterogeneous graph, denoted as $G = (U, V, E)$, encompasses two node types: $U = \{u_1, u_2, \dots, u_n\}$ for sample nodes (with n as the number of samples) and $V = \{v_1, v_2, \dots, v_k\}$ for a set of gene nodes (under one specific gene type, such as mRNA/DNA meth/miRNA, with k indicating the number of gene nodes). The connections between different node types in the heterogeneous graph are represented by $E = \{E_{u-u}, E_{u-v}, E_{v-v}\}$, where E_{u-u} , E_{u-v} , E_{v-v} respectively signify three types of edges: sample–sample, gene–gene, and sample–gene connections. Simultaneously, employing X_{sam} to denote the feature matrix of sample nodes in the heterogeneous network (where $X_{sam} \in \mathbb{R}^{n \times k}$, with n representing the number of samples and k the number of genes) and X_{fea} to represent the feature matrix of genes in the heterogeneous network ($X_{fea} \in \mathbb{R}^{k \times n}$). The construction process for the three types of edges in the heterogeneous graph is outlined as follows:

Sample–sample similarity network construction: Utilizing cosine similarity, the similarity A_{ij} between sample i and sample j is computed according to the following formula:

$$A_{ij} = S(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\|_2 \cdot \|x_j\|_2}, \quad (1)$$

where x_i, x_j denote the eigenvectors of sample i and sample j respectively ($x_i, x_j \in X_{sam}$), $x_i \cdot x_j$ denotes the inner product of the two vectors and $\|\cdot\|_2$ denotes the 2-parameter of the vector. After obtaining the sample similarity matrix $A \in \mathbb{R}^{n \times n}$, the edge matrix M_{ij} of the sample similarity network can be computed as:

$$M_{ij} = \begin{cases} 1, & \text{if } A_{ij} \geq \epsilon \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where ϵ is a threshold parameter whose main role is to control the number of edges in the similarity network, and whose value depends mainly on the average number of edges per node z :

$$z = \sum_{ij} I(A_{ij} > \epsilon) / n, \quad (3)$$

where $I(\cdot)$ is an indicator function, and n represents the number of nodes, i.e., the total number of samples. The parameter z is a tunable parameter, and the same value of z is employed consistently across all experiments on the same dataset. It is noteworthy that nodes in the sample similarity network include self-loops; when $z = 1$, there are no edges between any nodes.

Gene co-expression network construction: Weighted Gene Co-expression Network Analysis (WGCNA) is employed to identify clusters (modules) of genes highly correlated in expression characteristics, and it is widely applied in research areas such as the analysis of phenotype-genotype associations (Langfelder & Horvath, 2008). The co-expression matrix is constructed using the “WGCNA” package in the R programming language (Langfelder & Horvath, 2008). Using the gene matrix X_{fea} as input: the specific procedure for constructing the gene co-expression network with WGCNA is as follows:

Letting g_i and g_j represent the feature vectors of gene i and gene j ($g_i, g_j \in X_{X_{fea}}$), the correlation between the two genes can be calculated as:

$$s_{ij} = |cor(g_i, g_j)| = \left| \frac{\sum_{m=1}^n (g_{im} - \bar{g}_i)(g_{jm} - \bar{g}_j)}{\sqrt{\sum_{m=1}^n (g_{im} - \bar{g}_i)^2} \sqrt{\sum_{m=1}^n (g_{jm} - \bar{g}_j)^2}} \right|, \quad (4)$$

where $cor(\cdot)$ is the correlation calculation function, $|\cdot|$ denotes taking the absolute value of the resulting value, g_{im}, g_{jm} represent the m th feature of the feature vectors of genes i and j , $\bar{g}_i, \bar{g}_j$ are the mean of the two vectors, and n is the length of the vectors. After obtaining the gene similarity matrix, the co-expression adjacency matrix is generated by utilizing the adjacency function to convert the gene expression similarity matrix. The specific procedure is as follows:

$$P = [p_{ij}] = pow(s_{ij}, \beta) = |s_{ij}|^\beta, \quad (5)$$

where $pow(\cdot)$ represents the power function, β is a power-exponent hyperparameter threshold. To reduce noise and false correlations in the gene adjacency matrix, the co-expression adjacency matrix is transformed into a topological overlap matrix to obtain a new distance matrix:

$$\Omega = [\Omega_{ij}] = \frac{\sum_u p_{iu} p_{uj} + p_{ij}}{\min\{\sum_u p_{iu}, \sum_u p_{uj}\} + 1 - p_{ij}}, \quad (6)$$

where u represents all neighboring nodes of node i/j (excluding i/j), and $\sum_u p_{iu} p_{uj}$ represents the product of the co-expression values between node i/j and all nodes connected to both i and j , except i/j . The topological overlap matrix (TOM) takes into account the contribution of neighboring nodes to the co-expression relationship between pairs of genes. Following the determination of the topological overlap matrix for nodes, the representation of the edge matrix T_{ij} for the gene co-expression network is as follows:

$$T_{ij} = \begin{cases} 1, & \text{if } \Omega_{ij} \geq \lambda \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where Ω_{ij} is the topological overlap matrix, and λ is a threshold parameter primarily tasked with controlling the number of edges in the gene co-expression network.

Sample–gene association network construction: The sample–gene network is constructed based on the gene expression values of samples. Using the sample expression matrix X_{sam} as input and setting a threshold to control the edges between samples and genes, the edge matrix G_{ij} of the sample–gene association network can be obtained:

$$G_{ij} = \begin{cases} 1, & \text{if } X_{sam} \geq \eta \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where η is the threshold set to determine the edges in the network. After constructing the three subnetworks within the heterogeneous graph, the representation of the adjacency matrix E for the heterogeneous graph is as follows:

$$E = \begin{bmatrix} M & G^T \\ G & T \end{bmatrix} \in \mathbb{R}^{(n+k) \times (n+k)}, \quad (9)$$

where M , G , and T are the edge matrices for the sample similarity network, gene co-expression network, and sample–gene association network, respectively.

3.3. Omics-specific feature extraction based on HGCNs

After constructing the heterogeneous graph representing sample–gene interactions, the subsequent challenge lies in extracting high-order structural information from the graph. In this work, we have designed Heterogeneous Graph Convolutional Modules (HGCN) on the heterogeneous graph based on the message-passing strategy of GCN to effectively learn embedding representations for samples or genes. It is crucial to acknowledge the existence of two distinct node types in the graph, necessitating distinct processing approaches. To facilitate clarity, we initially focus on a single node type (e.g., sample nodes, denoted as U), subsequently extending our methodology to encompass the entire graph. In adhering to the message-passing paradigm (Gilmer et al., 2017), each GCN layer updates the embedding of every node based on the messages transmitted by its neighbors.

Let $H^l = \{h_{u_1}^l, \dots, h_{u_N}^l\}$ denote the embedding representation for all nodes at layer l (with l representing the layer and N representing the

number of nodes), and let $h_{u_i}^l$ be the input embedding for sample node $u_i (u_i \in U)$ at the l th layer. Thus, the output embedding for sample node u_i at the $l+1$ th layer can be formulated as follows:

$$h_{u_i}^{l+1} = \text{Update}(h_{u_i}^l, \text{Agg}(h_j^l | j \in N_i)), \quad (10)$$

where N_i represents the first-order neighbor set of node u_i , $\text{Agg}(\cdot)$ denotes the aggregation function for neighbor node embeddings, and $\text{Update}(\cdot)$ serves as the update function. The update function generates a new node embedding vector based on the previous node embedding vector and the aggregated result of its neighbors. For the embedding of sample node u_i , its aggregation encompasses information from neighboring sample nodes as well as associated gene nodes. To better capture distinctions between the aggregation of information from different node types, relation-specific weight matrices are introduced. Consequently, the output embedding for node u_i can be further delineated as:

$$h_{u_i}^{l+1} = \sigma(W_0^l h_{u_i}^l + \sum_{j \in N_{iu}} \frac{1}{c_{iu}} W_{u \rightarrow u}^l h_j^l + \sum_{g \in N_{iv}} \frac{1}{c_{iv}} W_{v \rightarrow u}^l h_g^l), \quad (11)$$

where N_{iu} and N_{iv} denote the set of neighboring sample nodes and neighboring gene nodes of node u_i , respectively; $h_{u_i}^l$, h_j^l and h_g^l denote the embedding of the sample node u_i in the l th layer, the embedding of the sample neighbor node j of u_i , and the embedding of the gene neighbor node g of u_i , respectively; the variables W_0^l , $W_{u \rightarrow u}^l$ and $W_{v \rightarrow u}^l$ represent three learnable transformation matrices corresponding to the fusion of weights from the previous layer, the weights for aggregating sample nodes, and the weights for aggregating gene nodes, respectively. c_{iu} and c_{iv} are normalization constants, which can be pre-selected (e.g., $c_{iu} = |N_{iu}|$, where $|N_{iu}|$ represents the number of connected samples to node i); $\sigma(\cdot)$ denotes the activation function. Similarly, for the update embedding of gene node v_i , there is formulated:

$$h_{v_i}^{l+1} = \sigma(W_0^l h_{v_i}^l + \sum_{j \in N_{iv}} \frac{1}{c_{iv}} W_{v \rightarrow v}^l h_j^l + \sum_{g \in N_{iu}} \frac{1}{c_{iu}} W_{u \rightarrow v}^l h_g^l), \quad (12)$$

where $h_{v_i}^l$ denotes the embedding of gene node v in the l th layer, and N_{iv} and N_{iu} denote the set of neighboring gene nodes as well as the set of neighboring sample nodes of node v_i , respectively. It is noteworthy that in order to make the sample nodes and gene nodes merge more efficiently, before performing the HGCN information aggregation, we perform a simple transformation of the node and gene features: $f = (WX + b)$ so that the two kinds of nodes are in the same dimensional space.

The above represent the feature updating process of HGCN. To broaden the application of HGCN to supervised learning, we first construct sample-gene heterogeneous networks $E_{tr}^{(i)} \in \mathbb{R}^{(n+k) \times (n+k)}$ for different omics training data $X_{tr}^{(i)}$ (where i denotes the respective omics type) using Eq. (9). Then, by using $X_{tr}^{(i)}$ and $E_{tr}^{(i)}$ as inputs to the graph attention network, the prediction for the training data can be formulated:

$$\hat{Y}_{tr}^{(i)} = \sigma(W(\text{HGCN}(X_{tr}^{(i)}, E_{tr}^{(i)})) + b), \quad (13)$$

where $\sigma(\cdot)$ is the activation function, W and b are the weights and paranoia of classified networks, $\hat{Y}_{tr}^{(i)} \in \mathbb{R}^{n \times c}$, the m th row in $\hat{Y}_{tr}^{(i)}$ denotes the estimated probabilities of labels for the m th training sample. Here, n signifies the total number of samples within the training dataset, while c denotes the number of classes involved in the classification task. $\text{HGCN}(\cdot)$ signifies the HGCN feature aggregation operation, employed for conducting the prediction task using the acquired features. Meanwhile, for the test set, each input involves a single test sample, forming the input feature matrix $X_{trte}^{(i)} \in \mathbb{R}^{(n+1) \times d}$ (with d representing the sample feature dimension). Based on Eq. (9), new nodes are added to construct the edge matrix $E_{trte}^{(i)} \in \mathbb{R}^{(n+1+k) \times (n+1+k)}$ for the sample-gene heterogeneous network. It is worth noting that the gene co-expression network portion of the heterogeneous graph in the test set remains consistent with that of the training set. Finally, by utilizing $\hat{Y}_{trte}^{(i)} = \text{HGCN}(X_{trte}^{(i)}, E_{trte}^{(i)})$, feature aggregation is performed, allowing the extraction of sample-specific features across different omics types. The

loss functions for the HGCNs under different omics types are defined as follows:

$$L_{HGCN}^{(i)} = \sum_{j=1}^{n_{tr}} L_{CE}(\hat{Y}_{tr}^{(i)}, Y_{tr}^i) = \sum_{j=1}^{n_{tr}} -\log\left(\frac{e^{\hat{y}_{j,c}^{(i)} \cdot y_j}}{\sum_c e^{\hat{y}_{j,c}^{(i)}}}\right), \quad (14)$$

where $L_{CE}(\cdot)$ denotes the cross-entropy loss function. $y_j \in \mathbb{R}^c$ corresponds to the one-hot encoded label for the j th training sample, with $\hat{y}_{j,c}^{(i)}$ representing the c th element within the vector $\hat{y}_j^{(i)}$, signifying the probability that the j th training sample is predicted to belong to class c . The superscript i indicates the i th omics type. Additionally, we introduced loss weights for different disease type to deal with the imbalanced training samples. Specifically, the loss weight for a particular class set as the reciprocal of the count of training data samples belonging to that class. The parameters trained in HGCNs enable feature aggregation across different omics types, uncovering unique characteristics of samples in various omics type spaces.

3.4. Confidence learning module for omics features

For multi-omics data classification tasks, the amount of information that different samples convey for classification in various omics types is often not fixed. Obtaining the variation in information content of samples across different omics types can effectively enhance the model's generalization ability. Simultaneously, it aligns well with biological realities, improving the model's biological interpretability. Therefore, we introduce an adaptive confidence-weighting technique based on True-Class-Probability (TCP) (Corbière et al., 2019) to assess the confidence scores of sample classifications in diverse omics types. These confidence scores are then used to update sample features.

Specifically, to obtain confidence scores for sample classification on different omics types, a separate classifier $f^m : x_n^m \rightarrow y_n$ is established for each omics type. For omics type m , the separate classifier f_m can be viewed as a probabilistic model responsible for transforming input values x^m into a predictive distribution $p^m(y|x^m) = [p_1^m, \dots, p_c^m]$, which is subsequently output through a Softmax operation. This classifier is trained using the cross-entropy loss function, with the objective of minimizing the distributional difference between predictions and labels:

$$L^{ls} = - \sum_{c=1}^C y_{ic} \log p_c^m, \quad (15)$$

where c represents the count of predicted class labels, and y_{ic} signifies the c th element within the class label y_i for sample i . Therefore, a commonly used measure for predicting confidence, known as the Maximum Class Probability (MCP), can be expressed as:

$$MCP^m = \max(p_1^m, \dots, p_c^m). \quad (16)$$

While Maximum Class Probability (MCP) has demonstrated strong performance in classification tasks, it can sometimes lead to overconfident predictions, especially in cases of misclassification (Moon, Kim, Shin, & Hwang, 2020; Van Amersfoort, Smith, Teh, & Gal, 2020). This is because the Softmax method used in MCP tends to widen the gap between the predicted probability and other probabilities. When the model incorrectly classifies a sample, the relevant probability for the true class becomes closer to a lower value. In order to mitigate instances of excessive confidence in predictions, TCP is introduced to yield predictions that are more dependable. Unlike MCP, where confidence is determined by selecting the maximum probability from the Softmax operation, TCP integrates the Softmax output probability associated with the true label as the confidence score. For instance, concerning a sample denoted as x within the context of omics m , where its predicted distribution $p^m(y|x^m) = [p_1^m, \dots, p_c^m]$ and the corresponding ground truth labels y are known, TCP^m (True-Class-Probability) can be expressed as follows:

$$TCP^m = y \cdot p^m(y|x^m) = \sum_{c=1}^C y_c p_c^m, \quad (17)$$

where $(\cdot)$ is a vector inner product operation. It is relatively straightforward to infer from the formula that when the model correctly classifies a sample, TCP aligns with MCP. However, in cases of misclassification by the model, TCP tends to approximate a lower confidence value, which often better reflects the fact of model misclassification.

However, it is important to note a critical drawback of TCP: it relies on the availability of ground truth labels for computation. When dealing with a test dataset, the true labels cannot be fed into the model for training purposes. To address this limitation, a confidence neural network, denoted as $c^m(x^m, \theta) \rightarrow TCP^m$ (θ represents the parameters of the neural network), is introduced. This network continually trains to approximate TCP^m using input features x^m . Due to the $TCP^m \in (0, 1)$, a sigmoid activation function is applied in the neural network's final layer, and the confidence neural network is trained using L2 loss:

$$L^{conf} = \sum_{m=1}^M (T\hat{C}P^m - TCP^m)^2 + L^{cls}, \quad (18)$$

where $T\hat{C}P^m = c^m(x^m, \theta)$ and M is the total number of omics. Once the confidence score $T\hat{C}P^m$ is obtained, it is used as attention weights to update the corresponding omics data as follows: $x^m = T\hat{C}P^m \cdot x^m$ for subsequent operations.

3.5. Cross modal-attention for multi-omics data fusion

Mining the potential relationship of multi-omics data can make better use of the complementary information between multi-omics data (Eddy, Mariani, & Kretzler, 2020). The self-attention mechanism is a method that maps different positions of a single sequence to one another, allowing the model to explore potential relationships among input vectors, facilitating the integration of information from other vectors as much as possible (Niu, Zhong, & Yu, 2021). Building upon this concept, MOHGCN has been designed with a cross-modal attention mechanism based on multi-head self-attention, enabling the comprehensive fusion of distinct omics-specific features. This approach thoroughly considers the interdependencies among various types of omics data, computes attention scores for different omics types, and ultimately combines multiple omics data through concatenation. In the end, sample classification is achieved by inputting the fused multi-omics data into a fully connected neural network. The specific operations are as follows:

To begin with, let the feature vector ensemble of sample a under various omics be denoted as $X_a = [x_a^1, \dots, x_a^{(m)}]$, where m signifies the number of omics types, and $x_a^{(m)}$ represents the specific feature of a sample within the m omics domain. For any given sample's collection of multi-omics feature vectors, three weight matrices W^q , W^k , and W^v are employed. These matrices are utilized to compute the Q , K , V values within the self-attention mechanism, with the calculation formula as follows:

$$Q = [q^1, \dots, q^{(i)}] = W^q \cdot X_a = W^q \cdot [x_a^1, \dots, x_a^{(m)}], \quad (19)$$

$$K = [k^1, \dots, k^{(i)}] = W^k \cdot X_a = W^k \cdot [x_a^1, \dots, x_a^{(m)}], \quad (20)$$

$$V = [v^1, \dots, v^{(i)}] = W^v \cdot X_a = W^v \cdot [x_a^1, \dots, x_a^{(m)}]. \quad (21)$$

The attention scores for each omics type can be calculated using the $\frac{QK^T}{\sqrt{d_k}}$ method within the self-attention mechanism. By multiplying Q and K and then dividing by the dimension of the input features, you can obtain the attention scores for each omics domain. Finally, normalize the obtained attention score matrix through the softmax function to obtain the attention score matrix A :

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right). \quad (22)$$

After obtaining the attention score matrix A , the next step involves further combining this attention score matrix with the value matrix V to obtain the output features $\hat{X}_a$. This specific operation can be represented using Eq. (23):

$$\hat{X}_a = AV = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right). \quad (23)$$

After completing the computation of single-head self-attention, this section employs a multi-head self-attention mechanism. We calculate multiple sets of weight matrices $W^{q,h}$, $W^{k,h}$, and $W^{v,h}$ for different branches (h is the number of multi-head self-attention heads.), enabling us to explore the inherent relationships among input features from various perspectives. Finally, by averaging the output results from multiple branches, we obtain the fused multi-omics features $\tilde{X}^{(m)}$ for sample a . We concatenate the features obtained through self-attention from various omics types $[\tilde{X}_a^{(1)}, \dots, \tilde{X}_a^{(m)}]$ and feed them into a fully connected neural network denoted as:

$$y_a = \sigma(W[\tilde{X}_a^{(1)}, \dots, \tilde{X}_a^{(m)}] + b). \quad (24)$$

This process allows us to obtain the predicted label y_a for the sample. The fully connected neural network utilizes cross-entropy loss for training, where $L^{ce} = -\sum_{c=1}^C y_{ac} \log p_{ac}$. Therefore, the loss for this section can be adjusted as follows:

$$L_{overall} = \sum_{i=1}^N (L^{conf} + L^{ce}). \quad (25)$$

In summary, in experiments involving m different types of omics data, MOHGCN utilizes a total loss function as follows:

$$L = \sum_{i=1}^m L_{HGCN}^{(i)} + L_{overall}. \quad (26)$$

During the training process of MOHGCN, it follows a two-step procedure. Initially, it pre-trains individual HGCNs for each omics type to obtain well-initialized HGCNs. Subsequently, within one epoch of the training process, it begins by updating the parameters of each omics-specific HGCN. Afterward, it freezes the omics-specific HGCNs and proceeds to update the parameters of the subsequent components, including TCP, cross modal-attention, and the fully connected neural network, to minimize the loss.

The framework MOHGCN operates through the following steps: Initially, for each omics dataset, we construct an omics-specific sample-gene heterogeneous graph based on our heterogeneous graph construction method. The constructed heterogeneous graphs and the associated sample-gene features are then fed into our designed omics-specific Heterogeneous Graph Convolutional Networks (HGCNs) to extract the omics-specific features. After extracting features, we input the sample features into the Confidence Learning Module (TCP). This module further refines and enhances the features to boost their classification capability. Subsequently, the features from the three different omics datasets are input into a self-attention-based cross-modal feature fusion module for feature fusion. The fused features are then passed through a fully connected layer to perform sample classification and prediction. The pseudo-code of the algorithmic flow of MOHGCN is shown in Algorithm 1. The code and data for MOHGCN are available at <https://github.com/ubuntu1024/MOHGCN>.

4. Materials

4.1. Dataset

We conducted extensive experiments on four publicly available datasets: ROSMAP (Alzheimer's Disease dataset), BRCA (Breast Cancer dataset), LGG (Low-Grade Glioma dataset), and KIPAN (Kidney Cancer dataset).

Table 1
Summary of datasets.

Dataset	Categories	Number of raw features	Number of features used for training
ROSMAP	NC: 169, AD: 182	mRNA:55889 meth:23788 miRNA:309	mRNA:200 meth:200 miRNA:200
BRCA	Normal-like: 115 Basal-like: 131, HER2-enriched: 46 Luminal A: 436, Luminal B: 147	mRNA:20531 meth:20106 miRNA:503	mRNA:1000 meth:1000 miRNA:503
LGG	Grade 2: 246, Grade 3: 264	mRNA:20531 meth:20114 miRNA:548	mRNA:2000 meth:2000 miRNA:548
KIPAN	KICH: 66 KIRC: 318 KIRP: 274	mRNA:20531 meth:20111 miRNA:445	mRNA:2000 meth:2000 miRNA:445

Algorithm 1 Algorithm of MOHGCN

Input: Number of omics m , feature matrix of different omics data X^m , true label y , hyper-parameters ϵ, λ, η , training epochs $epoch$.

Output: Predicted labels y_a .

```

1: Initialize feature matrices  $X_{sam}^m, X_{fea}^m$  from  $X^m$ ;
2: Construct heterogeneous graph  $E^m$  for each omics using  $\epsilon, \lambda, \eta$  with
   Eq. (2), Eq. (7), Eq. (8), and Eq. (9);
3: for  $n = 1$  to  $epoch$  do
4:   for  $i = 1$  to  $m$  do
5:     Update node features using  $HGCN^{(i)}$  with inputs  $X_{sam}^m, X_{fea}^m$ 
     and  $E^m$  according to Eq. (11) and Eq. (12);
6:     Calculate the  $TCP_m$  according to Eq. (17);
7:     Update sample features with  $x^m = T\hat{C}P^m \cdot x^m$ ;
8:     Compute the loss  $L_{HGCN}^{(i)}$  and  $L^{conf}$  in turn with Eq. (14) and
     Eq. (18);
9:   Minimize  $L_{HGCN}^{(i)}$  for  $HGCN^{(i)}$  training;
10:  end for
11:  Calculate  $Q, K$ , and  $V$  and update sample features  $X$  with
   Eq. (19), Eq. (20), Eq. (21), and Eq. (23);
12:  Calculate predicted labels  $y_a$  according to Eq. (24);
13:  Calculate  $L_{overall}$  according to Eq. (25);
14:  Update TCP layer, cross-modal attention layer, and fully
   connected layer by minimizing  $L_{overall}$ ;
15: end for
16: return  $y_a$ 

```

Specifically: ROSMAP dataset: Used for classifying two classes of Alzheimer's Disease (AD) patients. BRCA dataset: Employed for classifying five subtypes of breast cancer according to PAM50 classification. LGG dataset: Utilized for classifying two grades of low-grade glioma. KIPAN dataset: Employed for classifying three types of kidney cancer. In all of these datasets, we used three types of omics data: mRNA expression data (mRNA), miRNA expression data (miRNA), and DNA methylation data (meth). Table 1 provides detailed information about these datasets. For further details on data collection and preprocessing, please refer to Wang et al. (2021).

4.2. Compared methods

In order to fully investigate the recent state of the art in multi-omics data fusion research, we compare a total of 15 existing computational approaches, including six single-omics classifiers using early integrated strategies (SVM (Lanckriet et al., 2004), KNN (Fix & Hodges, 1951), Lasso regression-based (Ranstan & Cook, 2018), random forest (RF) (Ho, 1995), fully connected neural networks (FCNN), and extreme gradient boosting (XGboost) (Chen & Guestrin, 2016)), and nine state-of-the-art methods including Block Partial Least Squares Discriminant

Analysis (BPLSDA) (Singh et al., 2019) involves the exploration of multimodal data within the latent space through the means of discriminant analysis. Block Sparse Partial Least Squares Discriminant Analysis (BSPLSDA) (Singh et al., 2019) employs sparse constraints within BPLSDA to select the most relevant features. Group regularized (logistic) ridge regression (GRidge) (Van De Wiel et al., 2016) is employed for the structured utilization of multimodal data, utilizing group-specific penalties. Cascade of Final Multimodal Representations (CF) (Hong et al., 2020) accomplishes the integration of multiple modalities through the sequential cascading of late multimodal representations. Trusted Multiview Classification (TMC) (Han, Zhang, Fu and Zhou, 2022) executes decision fusion grounded in the confidence levels associated with various modalities. Gated Multimodal Unit (GMU) (Arevalo, Solorio, Montes-y Gomez, & González, 2020) facilitates information fusion by constructing intermediate multimodal representations through the combination of data. Mogonet (Wang et al., 2021) constructs a graph for each modality and explores multimodal correlations through a correlation discovery network. Dynamic (Han, Yang et al., 2022) dynamically evaluates the informativeness of different samples at both the feature level and the modality level, thus integrating multiple modalities. MLCLNet (Zheng, Liu et al., 2023) integrates multiple modalities using the feature confidence learning mechanism with graph convolutional neural networks for feature extraction and uses a cross-modal label fusion module for multimodal fusion. DPNET (Zou et al., 2023) is a dynamic multi-attention network that integrates the modeling of multi-omics data and label confidence to enhance the stability and trustworthiness of classification fusion. HyperTMO (Wang et al., 2024) employs a hypergraph convolutional network to learn higher-order relationships in multi-omics data and integrates these multi-omics datasets for disease classification.

4.3. Performance assessment

In order to assess the effectiveness of the MOHGCN model proposed in this paper, classification evaluation metrics are introduced for performance evaluation. For binary classification tasks, the following metrics are employed: accuracy (ACC), F1-score (F1), and AUC. For multi-class classification tasks, the evaluation includes accuracy (ACC), weighted F1-score (F1_weighted), and macro-averaged F1-score (F1_macro) among other metrics, to comprehensively assess the model's performance in multi-class classification scenarios. The details of the evaluation metrics used in the experiments are consistent with those described in Mogonet (Wang et al., 2021).

The optimization problem for training the model focuses on minimizing the training loss to achieve superior classification performance. This involves two key aspects: optimizing the loss $L_{HGCN}^{(i)}$ for extracting specific features from each omics data type to enhance the model's feature extraction capability, and optimizing $L_{overall}$ to strengthen feature fusion and classification performance. The network training utilizes the

Table 2

Various methodologies for binary classification outcomes on the ROSMAP and LGG datasets.

Method	ROSMAP			LGG		
	ACC	F1	AUC	ACC	F1	AUC
SVM	0.770 ± 0.024	0.778 ± 0.016	0.770 ± 0.026	0.754 ± 0.046	0.757 ± 0.050	0.754 ± 0.046
KNN	0.657 ± 0.036	0.671 ± 0.045	0.709 ± 0.045	0.729 ± 0.034	0.738 ± 0.033	0.799 ± 0.038
Lasso	0.694 ± 0.037	0.730 ± 0.033	0.770 ± 0.035	0.761 ± 0.018	0.767 ± 0.022	0.823 ± 0.027
RF	0.726 ± 0.029	0.734 ± 0.021	0.811 ± 0.019	0.748 ± 0.012	0.742 ± 0.010	0.823 ± 0.010
XGboost	0.760 ± 0.046	0.772 ± 0.045	0.837 ± 0.030	0.756 ± 0.040	0.767 ± 0.032	0.840 ± 0.023
FCNN	0.755 ± 0.021	0.764 ± 0.021	0.827 ± 0.025	0.737 ± 0.023	0.748 ± 0.024	0.810 ± 0.037
BPLSDA	0.742 ± 0.024	0.755 ± 0.023	0.830 ± 0.025	0.759 ± 0.025	0.738 ± 0.031	0.825 ± 0.023
BSPLSDA	0.753 ± 0.033	0.764 ± 0.035	0.838 ± 0.021	0.685 ± 0.025	0.662 ± 0.030	0.730 ± 0.026
GRidge	0.760 ± 0.034	0.769 ± 0.029	0.841 ± 0.023	0.746 ± 0.038	0.756 ± 0.036	0.826 ± 0.044
CF	0.784 ± 0.011	0.788 ± 0.005	0.880 ± 0.005	0.811 ± 0.012	0.822 ± 0.004	0.881 ± 0.004
TMC	0.825 ± 0.009	0.823 ± 0.006	0.885 ± 0.006	0.819 ± 0.008	0.815 ± 0.004	0.871 ± 0.004
GMU	0.776 ± 0.025	0.784 ± 0.016	0.869 ± 0.016	0.803 ± 0.015	0.808 ± 0.012	0.886 ± 0.012
Mogonet	0.815 ± 0.023	0.821 ± 0.022	0.874 ± 0.012	0.816 ± 0.016	0.814 ± 0.027	0.840 ± 0.027
Dynamic	0.842 ± 0.013	0.846 ± 0.007	0.912 ± 0.007	0.833 ± 0.010	0.837 ± 0.004	0.885 ± 0.004
MLCLNet	0.844 ± 0.015	0.852 ± 0.015	0.893 ± 0.011	0.835 ± 0.014	0.840 ± 0.013	0.886 ± 0.012
DPNET	0.851 ± 0.011	0.848 ± 0.007	0.913 ± 0.007	0.838 ± 0.013	0.847 ± 0.012	0.889 ± 0.011
HyperTMO	0.875 ± 0.033	0.874 ± 0.033	0.900 ± 0.030	0.826 ± 0.017	0.824 ± 0.013	0.881 ± 0.007
Ours	0.881 ± 0.005	0.885 ± 0.005	0.907 ± 0.006	0.846 ± 0.005	0.855 ± 0.006	0.892 ± 0.006

Table 3

Various methodologies for multi-classification outcomes on the BRCA and KIPAN datasets.

Method	BRCA			KIPAN		
	ACC	F1_weighted	F1_macro	ACC	F1_weighted	F1_macro
SVM	0.729 ± 0.018	0.702 ± 0.015	0.640 ± 0.017	0.995 ± 0.003	0.995 ± 0.003	0.994 ± 0.004
KNN	0.742 ± 0.024	0.730 ± 0.023	0.682 ± 0.025	0.967 ± 0.011	0.967 ± 0.011	0.960 ± 0.014
Lasso	0.732 ± 0.012	0.698 ± 0.015	0.642 ± 0.026	0.974 ± 0.002	0.974 ± 0.002	0.972 ± 0.004
RF	0.754 ± 0.009	0.733 ± 0.010	0.649 ± 0.013	0.981 ± 0.006	0.981 ± 0.006	0.975 ± 0.011
XGboost	0.781 ± 0.008	0.764 ± 0.010	0.701 ± 0.017	0.993 ± 0.008	0.993 ± 0.008	0.989 ± 0.014
FCNN	0.754 ± 0.028	0.740 ± 0.034	0.668 ± 0.047	0.991 ± 0.005	0.991 ± 0.005	0.991 ± 0.005
BPLSDA	0.642 ± 0.009	0.534 ± 0.014	0.369 ± 0.017	0.933 ± 0.013	0.933 ± 0.013	0.919 ± 0.021
BSPLSDA	0.639 ± 0.008	0.522 ± 0.016	0.351 ± 0.022	0.919 ± 0.012	0.918 ± 0.013	0.895 ± 0.014
GRidge	0.745 ± 0.016	0.726 ± 0.019	0.656 ± 0.025	0.994 ± 0.004	0.994 ± 0.004	0.993 ± 0.004
CF	0.815 ± 0.008	0.815 ± 0.009	0.771 ± 0.009	0.992 ± 0.005	0.992 ± 0.005	0.988 ± 0.009
TMC	0.842 ± 0.005	0.844 ± 0.009	0.806 ± 0.009	0.997 ± 0.003	0.997 ± 0.003	0.994 ± 0.005
GMU	0.800 ± 0.039	0.798 ± 0.058	0.746 ± 0.058	0.977 ± 0.016	0.976 ± 0.017	0.958 ± 0.032
Mogonet	0.829 ± 0.018	0.825 ± 0.016	0.774 ± 0.017	0.999 ± 0.002	0.999 ± 0.002	0.999 ± 0.002
Dynamic	0.877 ± 0.003	0.880 ± 0.005	0.845 ± 0.005	0.999 ± 0.002	0.999 ± 0.002	0.999 ± 0.003
MLCLNet	0.864 ± 0.016	0.878 ± 0.015	0.826 ± 0.018	0.999 ± 0.003	0.999 ± 0.002	0.999 ± 0.002
DPNET	0.878 ± 0.010	0.884 ± 0.012	0.852 ± 0.012	0.999 ± 0.002	0.999 ± 0.002	0.999 ± 0.003
HyperTMO	0.858 ± 0.023	0.863 ± 0.023	0.841 ± 0.019	0.999 ± 0.002	0.999 ± 0.002	0.999 ± 0.002
Ours	0.883 ± 0.003	0.887 ± 0.002	0.864 ± 0.006	0.999 ± 0.003	0.999 ± 0.002	0.999 ± 0.002

Adam optimizer with a learning rate decay strategy. For each dataset in our experiments, training is halted at 1200 epochs, after which the test results are evaluated and reported.

5. Results

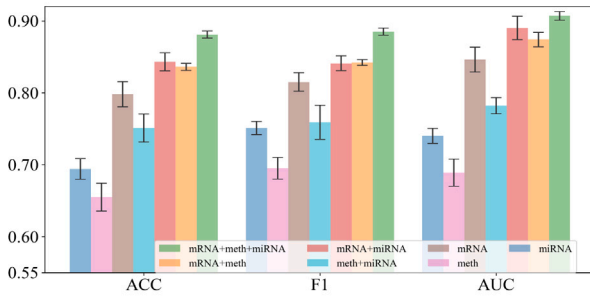
To conduct a comprehensive comparison of the classification performance between the proposed MOHGCN framework and other methods, experiments were carried out on four datasets, and comparisons were drawn against 15 other approaches. All methods were performed in a uniform setting. For each dataset, a random selection of 30% of the samples was allocated to the test set, with the remaining 70% reserved for the training set. The experimental results were obtained as the average values over 10 experiments, along with the corresponding standard deviations. The MOHGCN framework exhibited superior performance, as substantiated by comprehensive experiments conducted on diverse datasets. Additionally, a series of experiments, including model performance across different omics, functional module ablation experiments, ablation of the effect of heterogeneous diagrams transferring information, visualization of features before and after the model, further validated the effectiveness of the MOHGCN.

5.1. Complex disease patient classification performance

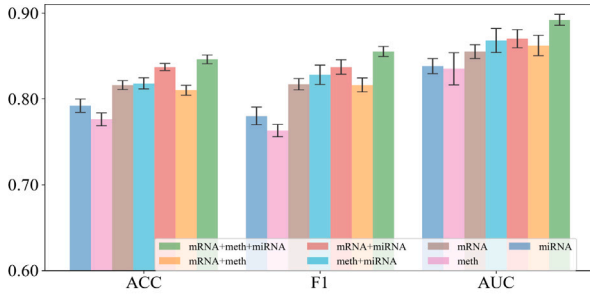
Binary Classification Performance Comparison: A comparison between the proposed MOHGCN framework and 15 other methods

for integrating multi-omics data was undertaken, utilizing two binary classification datasets, namely, the ROSMAP dataset and the LGG dataset, as detailed in Table 2. Model performance was assessed using binary classification metrics, including ACC, F1, and AUC. Table 2 clearly illustrates that, among all methods evaluated on both binary classification datasets, the proposed MOHGCN method delivers the highest performance with respect to ACC and F1 scores. It outperforms other methods significantly. Furthermore, with regards to the AUC metric, MOHGCN exhibits robust performance, showing only slight deviations when compared to a subset of methods. The experimental findings demonstrate the excellence of the proposed framework in binary classification tasks that rely on the integration of multi-omics data.

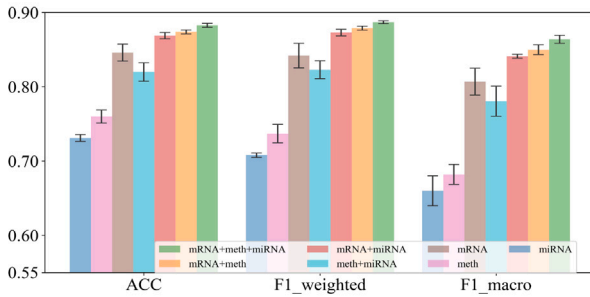
Multi-Class Classification Performance Comparison: The performance of the proposed MOHGCN framework was assessed in comparison with 15 other methods for multi-omics data integration, utilizing two multi-class classification datasets, namely BRCA (breast cancer dataset) and KIPAN (kidney cancer dataset), as presented in Table 3. The evaluation of model performance encompassed multi-class classification metrics, which included ACC, F1_weighted, and F1_macro. Table 3 provides clear evidence that, for the classification of five subtypes of breast cancer and three classes of kidney cancer, MOHGCN outperforms other methods across all evaluation metrics. The experimental findings serve to illustrate the robust performance of the MOHGCN framework in the context of multi-class classification tasks that rely on the integration of multi-omics data.



(a) The experimental result for various combinations of omics data on the ROSMAP dataset.



(b) The experimental result for various combinations of omics data on the LGG dataset.



(c) The experimental result for various combinations of omics data on the BRCA dataset.

Fig. 2. The impact of various combinations of omics data on the model's classification outcomes across the three datasets.

In order to visually represent the convergence of the proposed network during training, we plot the training loss and training accuracy of MOHGCN on the BRCA dataset with different training iteration epochs, as shown in Fig. 3. It can be seen that our proposed MOHGCN can converge well within 1000 iteration epochs.

5.2. Investigation of model performance across different combinations of omics

Some experiments were conducted to evaluate MOHGCN classification performance with different multi-omics data as well as the performance with individual omics data. The MOHGCN framework was used to explore the impact of combining different omics data on experiments across BRCA, ROSMA, and LGG datasets. Specifically, since a single omics cannot utilize self-attention for integrating different omics, experiments using only a single omics for classification were conducted with the Heterogeneous Graph Convolutional Network (HGCN) module and the TCP module combined with a fully connected neural network. The specific experimental results are shown in Fig. 2.

Experiments were conducted using three types of omics data (mRNA + DNA methylation + miRNA), two types of omics data (combinations

of mRNA + DNA methylation, mRNA + miRNA, DNA methylation + miRNA), and with experiments that utilized only a single type of omics data to train omics-specific HGCN modules and TCP modules. The results across the three datasets demonstrate that the best performance is achieved when employing methods that integrate information from three different modalities. Furthermore, by comparing the classification performance of individual omics experiments with the integration of multiple omics experiments, we observed a sustained enhancement in classification performance across the majority of datasets. The experimental results further suggest that complex diseases often occur due to interactions among multiple omics. Integrating different omics data provides complementary insights from various perspectives and enhances classification performance. This further emphasizes the significance of multi-omics data integration in bolstering classification performance within the realm of biomedical applications.

5.3. Experimental research on ablation of functional modules

To comprehensively understand the impact of the constituent modules of the MOHGCN framework on classification tasks, we conducted further ablation studies on the three critical components of MOHGCN across four datasets. Specifically, each of the three key components, namely, the omics-specific feature extraction modules (HGCNs), the feature assessment module (TCP), and Self-attention based cross-modal multi-omics data fusion module (called CMAtt module), were systematically ablated. These ablated models were then compared to the full MOHGCN approach. When ablating the HGCNs module, we ensured the uniform dimensionality of different omics data for subsequent multi-omics fusion by applying a single-layer linear transformation to the input omics features. To clearly illustrate the impact of module ablation on the results, One-way analysis of variance (ANOVA) (Kim, 2017) was conducted on each performance metric following the ablation of the modules. And p-values less than 0.05 indicate significant differences between groups. The results of the experiments are detailed in Table 4.

The experimental results reveal the following findings: (1) The removal of any module results in a decline in classification performance, with p-values from the ANOVA tests for different metrics being less than 0.05. This highlights significant differences following the ablation of various modules, indicating the crucial role of all modules within the MOHGCN model. (2) Notably, when the cross modal-attention (CMAtt) module was omitted, a reduction in performance was observed across most datasets. This observation implies that the cross modal-attention (CMAtt) module adeptly explores correlations between different omics types, resulting in more promising outcomes on the majority of datasets. (3) The MOHGCN framework exhibited superior performance on most datasets, owing to the comprehensive information extraction by the HGCNs module and the data fusion facilitated by the cross modal-attention (CMAtt) module. These findings underscore the critical importance of each module and affirm the overall effectiveness of the MOHGCN framework.

5.4. Ablation of information propagation in heterogeneous graphs

The design of the heterogeneous graph structure not only enhances the aggregation of information from neighboring sample nodes but also enables samples to propagate their information to adjacent nodes through self-associated gene features. This involves the transmission of information through associated genes, increasing model interpretability and simultaneously improving classification efficiency. To thoroughly investigate information propagation on the heterogeneous graph structure, we conducted ablation experiments on four datasets. Specifically, in the construction of the sample-gene association network, we set the threshold for constructing the sample-gene association network to 1, completely severing the connections between sample and gene nodes. This restriction allows sample nodes to only aggregate information from their neighboring nodes. Meanwhile, in order to compare the

Table 4
Ablation experiment of different components of MOHGCN.

Dataset	Model	ACC	F1_weighted	F1_macro
BRCA	MOHGCN_noHGCN	0.863 ± 0.006	0.866 ± 0.006	0.832 ± 0.011
	MOHGCN_noTCP	0.867 ± 0.006	0.872 ± 0.002	0.842 ± 0.002
	MOHGCN_noCMAtt	0.865 ± 0.004	0.866 ± 0.004	0.840 ± 0.004
	MOHGCN	0.883 ± 0.003	0.887 ± 0.002	0.864 ± 0.006
	<i>p-value</i>	2.86e-11	2.38e-12	9.12e-11
KIPAN	MOHGCN_noHGCN	0.984 ± 0.007	0.983 ± 0.007	0.980 ± 0.016
	MOHGCN_noTCP	0.992 ± 0.006	0.992 ± 0.006	0.992 ± 0.007
	MOHGCN_noCMAtt	0.997 ± 0.002	0.997 ± 0.002	0.996 ± 0.004
	MOHGCN	0.999 ± 0.003	0.999 ± 0.002	0.999 ± 0.002
	<i>p-value</i>	5.55e-7	1.92e-6	1.09e-3
Dataset	Model	ACC	F1	AUC
ROSMAP	MOHGCN_noHGCN	0.836 ± 0.004	0.839 ± 0.005	0.889 ± 0.003
	MOHGCN_noTCP	0.847 ± 0.011	0.848 ± 0.009	0.901 ± 0.009
	MOHGCN_noCMAtt	0.824 ± 0.012	0.831 ± 0.009	0.862 ± 0.023
	MOHGCN	0.881 ± 0.005	0.885 ± 0.005	0.907 ± 0.006
	<i>p-value</i>	8.49e-15	1.82e-16	5.79e-8
LGG	MOHGCN_noHGCN	0.822 ± 0.015	0.825 ± 0.013	0.884 ± 0.015
	MOHGCN_noTCP	0.838 ± 0.007	0.846 ± 0.011	0.885 ± 0.007
	MOHGCN_noCMAtt	0.832 ± 0.006	0.834 ± 0.011	0.879 ± 0.005
	MOHGCN	0.846 ± 0.005	0.855 ± 0.006	0.892 ± 0.006
	<i>p-value</i>	4.16e-5	6.25e-6	3.64e-2

Table 5
Experimental results of ablating heterogeneous graph structures on four datasets.

Dataset	Model	ACC	F1	AUC
ROSMAP	No_heter_graph	0.845 ± 0.012	0.840 ± 0.012	0.901 ± 0.018
	MOHGCN	0.881 ± 0.005	0.885 ± 0.005	0.907 ± 0.006
	<i>p-value</i>	2.49e-7	6.79e-9	7.76e-3
LGG	No_heter_graph	0.827 ± 0.002	0.822 ± 0.010	0.890 ± 0.005
	MOHGCN	0.846 ± 0.005	0.855 ± 0.006	0.892 ± 0.006
	<i>p-value</i>	6.08e-9	2.07e-7	3.77e-2
Dataset	Model	ACC	F1_weighted	F1_macro
BRCA	No_heter_graph	0.865 ± 0.004	0.869 ± 0.004	0.836 ± 0.010
	MOHGCN	0.883 ± 0.003	0.887 ± 0.002	0.864 ± 0.006
	<i>p-value</i>	5.59e-9	2.29e-9	2.42e-6
KIPAN	No_heter_graph	0.996 ± 0.003	0.996 ± 0.003	0.994 ± 0.008
	MOHGCN	0.999 ± 0.003	0.999 ± 0.002	0.999 ± 0.002
	<i>p-value</i>	2.77e-2	2.59e-2	1.79e-3

differences between the two groups of results before and after the ablation of information propagation in heterogeneous graphs, Student's t-test (Delacre, Lakens, & Leys, 2017) was utilized to analyze the performance of various metrics. The experimental results on the four datasets are presented in Table 5.

From the results in the Table 5, it is evident that, after ablating information propagation on the heterogeneous graph, there is a varying degree of decline in classification performance across the four datasets. Furthermore, the p-values from the t-tests across all metrics are less than 0.05, underscoring the significant differences between the groups before and after the ablation of information propagation in heterogeneous graphs. This indirectly validates the effectiveness of the heterogeneous graph design. The experimental results demonstrate that the mechanisms underlying complex diseases are often closely linked to intricate interactions among biomolecules across different omics. These complex biomolecular interactions within various omics enhance the classification performance of the model.

5.5. Visualization of features before and after the MOHGCN

To show the MOHGCN framework's ability to effectively extract and amalgamate latent features from multi-omics data, a comparative analysis was conducted on the BRCA dataset. This analysis involved comparing the original input data with the embeddings generated by applying MOHGCN to the samples, followed by visualizing the results.

Concretely, the original input data was created by concatenating the multi-omics data for each sample. Subsequently, MOHGCN was employed to process the multi-omics data, yielding latent embeddings representing the samples. For the purpose of visualization, t-distributed Stochastic Neighbor Embedding (t-SNE) (Van der Maaten & Hinton, 2008), a frequently employed dimensionality reduction technique for visualizing and interpreting high-dimensional data, was utilized. The experimental findings are illustrated in Fig. 4.

Upon meticulous examination, Fig. 4(a) clearly reveals that the samples in the original dataset are closely clustered, posing challenges in effectively discriminating among the five distinct types of breast cancer samples. Conversely, the sample feature embeddings obtained after undergoing MOHGCN processing facilitate more effective differentiation among different sample types. Notably, the graph conspicuously demonstrates that, with few exceptions, each type of sample cluster exhibits relatively greater distinctiveness compared to their clustering based on their original features. This observation underscores the MOHGCN framework's proficiency in processing and amalgamating multi-omics data, resulting in embeddings that can be effectively employed for classification tasks, thus demonstrating superior performance.

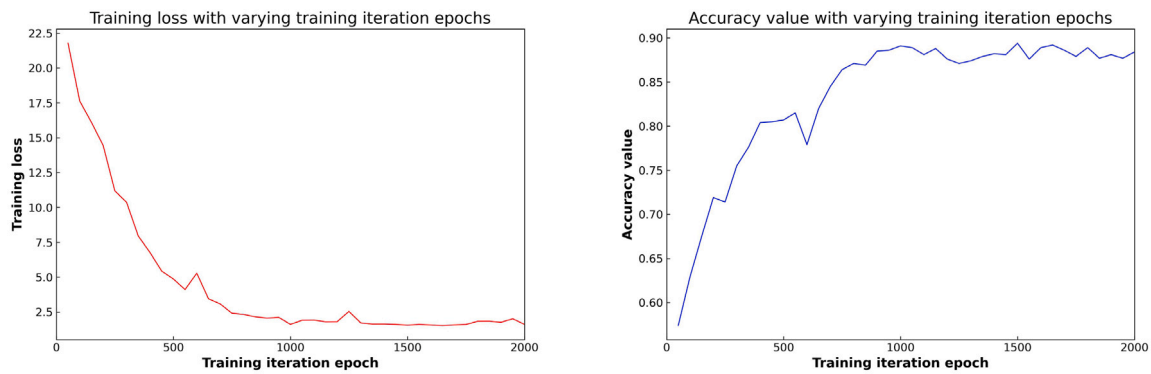
6. Conclusion

In this work, we propose a trustworthy multi-omics data integration framework called MOHGCN based on specificity-aware heterogeneous graph convolutional neural networks. The framework maximizes the interactions between biomolecules in patients under specific diseases, incorporating a heterogeneous graph convolution module. It employs trustworthy attention weighting and self-attention feature fusion to facilitate the extraction and integration of features from different omics data. This approach realizes the exploration of potential correlations between omics, enhancing the diagnosis of complex diseases while avoiding mutual interference among different omics datasets. Through extensive experiments on four public datasets, the proposed MOHGCN framework demonstrates superior performance compared to state-of-the-art multi-omics data integration methods.

Concurrently, experiments were conducted to assess the efficacy of MOHGCN across diverse omics integrations. The findings underscore the importance of integrating diverse omics data to enhance classification performance in biomedical applications. Additionally, diverse experiments, including functional module ablation, ablation of heterogeneous graph information propagation, and visualization of features before and after model application, highlight the superiority and effectiveness of the proposed MOHGCN framework.

7. Discussion

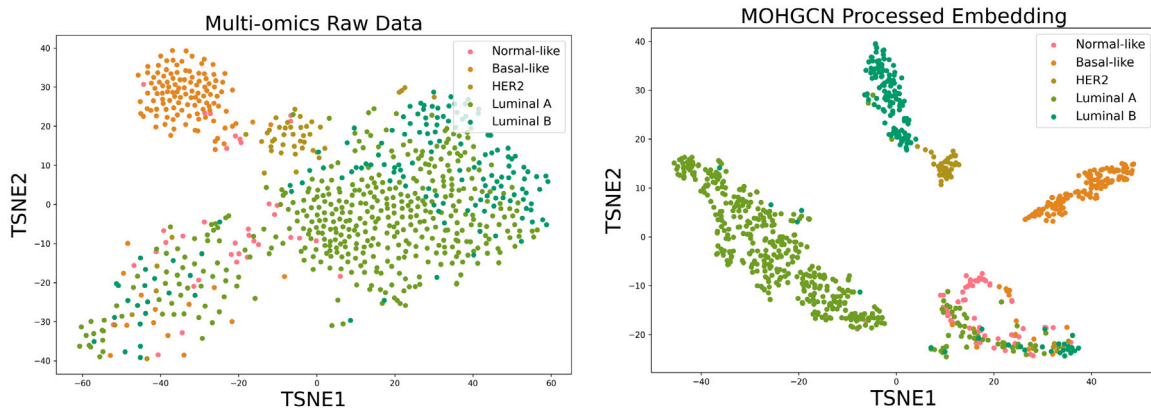
The anticipated robustness and interpretability of MOHGCN, in theory, are rooted in several key considerations. Initially, a meticulously crafted sample-gene heterogeneous network captures intricate biomolecular interactions within patients affected by specific diseases. The introduction of the Heterogeneous Graph Convolutional (HGCN) onto the heterogeneous graph plays a pivotal role in effectively extracting discerning features from samples, significantly enhancing the model's interpretability. On another front, the diverse branches of HGCN demonstrate commendable proficiency in extracting distinctive features corresponding to different omics. This strategic approach effectively addresses issues of imbalance during the model learning process, stemming from the disparate distribution of multi-omics data and the inherent heterogeneity among omics. Furthermore, the framework incorporates a self-attention mechanism to systematically uncover potential correlations within various omics data, facilitating the seamless fusion of multi-omics data. Rigorous ablation experiments highlight the pivotal role of the self-attention mechanism in the model architecture. At the same time, our proposed MOHGCN framework can be applied in a variety of fields including computer-aided diagnosis based on the



(a) The training loss with varying training iteration epochs.

(b) The accuracy values varying training iteration epochs.

Fig. 3. Training loss and accuracy values with varying training iteration epochs on BRCA dataset.



(a) The t-SNE visualization of the raw feature data.

(b) The t-SNE of the features post MOHGCN processing.

Fig. 4. TSNE diagram of the effect change before and after the model feature update on the BRCA dataset.

integration of multi-omics data, integrated analysis of multi-omics data, and other application areas.

Nevertheless, it is worth acknowledging that the proposed framework, MOHGCN, does exhibit certain limitations. Firstly, training a heterogeneous graph convolutional network (HGCN) for each omics results in longer runtimes compared to direct integration methods utilizing deep neural networks. Secondly, for diseases with extremely limited sample sizes, the framework's substantial number of parameters makes it prone to overfitting during training, which leads to difficulties in achieving satisfactory performance. In future work, a primary focus will be on reducing model complexity and researching diagnostic methodologies tailored for diseases with a limited number of samples.

CRedit authorship contribution statement

Wenhao Wu: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Shudong Wang:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition. **Yuanyuan Zhang:** Conceptualization, Writing – review & editing. **Kuijie Zhang:** Software, Validation, Visualization. **Wenjing Yin:** Software, Visualization. **Shanchen Pang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data/code in the manuscript.

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